



Abiotic conditions affect nectar properties and flower visitation in four herbaceous plant species

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ABSTRACT

Both plant nectar production and insect activity are highly dependent on abiotic environmental conditions. Furthermore, the foraging behaviour of insect pollinators can be affected by nectar properties. In the context of climate change, it is important to understand how plant-pollinator interactions respond to temperature and other abiotic factors. We investigated the effect of natural variation in temperature and solar radiation on nectar quantity (nectar volume) and quality (sucrose concentration and sucrose mass) and on flower visitation rates in four herbaceous plant species (*Dictamnus albus*, *Lamium album*, *Salvia officinalis*, *Vincetoxicum hirsutum*) in the Botanical Garden Halle (Germany). Temperature affected nectar properties in all four species. Solar radiation affected nectar quantity and quality in two species, most likely by affecting flower temperature. The number of flower visits was unimodally related to temperature for two species and positively related to solar radiation in another. The variable responses across plant species in the effects of abiotic factors on nectar properties and flower visitation patterns may be due to differences in flower shape and colour, to differences in the composition of flower visitors, or due to other unmeasured extrinsic factors that vary across patches where these species occur. Our study highlights the importance of considering direct and indirect effects of climate factors on pollinator visitation in multiple plant species.

1. Introduction

Pollination is amongst the most important ecosystem services for both natural and agricultural systems (Klein et al., 2007), as the majority of plant species require animal visitation in order to successfully reproduce (Ollerton et al., 2011; Rodger et al., 2021). Climate change has the potential to significantly threaten this critical interaction (González-Varo et al., 2013). Climate can directly affect insect flower visitors because insects have species-specific temperature optima for foraging (Corbet et al., 1993; Stone and Willmer, 1989; Zoller et al., 2020), and indirectly influence flower visitors by altering the rewards (e.g., nectar volume and sucrose amount) that plants offer (Descamps et al., 2018). To develop a general understanding of how climate change may alter pollination, it is important to understand how each of these

components will change in response to abiotic factors such as temperature and solar radiation (Maron et al., 2014), as these factors are likely to be affected by climate change (Rapp, 2014).

For a given community of potential flower visitors, the visitation rate and composition of visitor communities to flowering plants depends on the characteristics of the plant species. Plant species differ in their floral traits, such as the size, shape and colour of the corolla, scent and nectar composition, which are associated with their pollination syndrome (Faegri and Pijl, 1979; Rosas - Guerrero et al., 2014). These traits might differ significantly across even closely related plants (e.g. Basnett et al., 2021; Comba, 1999; Maćukanović-Jocić et al., 2011). For example, *Salvia officinalis* L. and *Lamium album* L. are both plant species in the Lamiaceae family that attract bees with their bilabiate corollas and nectar rewards. However, only bees with a long proboscis are able to

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